

PL/SQL Programming

Exercise 1: Control Structures

1. Table Creation

The screenshot shows the SQL Developer interface. The Navigator pane on the left displays 'My Schema' with tables 'CUSTOMERS' and 'LOANS'. The main editor window, titled '[SQL Worksheet]*', contains the following SQL code:

```
1 CREATE TABLE Customers (  
2   CustomerID NUMBER PRIMARY KEY,  
3   Age NUMBER,  
4   Balance NUMBER,  
5   IsVIP VARCHAR2(5)  
6 );  
7  
8 CREATE TABLE Loans (  
9   LoanID NUMBER PRIMARY KEY,  
10  CustomerID NUMBER,  
11  InterestRate NUMBER,  
12  DueDate DATE  
13 );
```

The 'Script output' tab is active, showing the execution results:

```
Age NUMBER,  
Balance NUMBER,...  
Show more...
```

Below the output, it states 'Table CUSTOMERS created.' The footer of the application shows '26.5.2 - Copyright © 2015, 2026 Oracle and/or its affiliates All rights reserved'.

This screenshot shows the same SQL Developer interface as the previous one, but with the second table created. The SQL code in the editor is:

```
1 CREATE TABLE Customers (  
2   CustomerID NUMBER PRIMARY KEY,  
3   Age NUMBER,  
4   Balance NUMBER,  
5   IsVIP VARCHAR2(5)  
6 );  
7  
8 CREATE TABLE Loans (  
9   LoanID NUMBER PRIMARY KEY,  
10  CustomerID NUMBER,  
11  InterestRate NUMBER,  
12  DueDate DATE  
13 );
```

The 'Script output' tab now shows the results for the second table:

```
Balance NUMBER,...  
Show more...
```

Below this, it states 'Table CUSTOMERS created.' followed by 'Elapsed: 00:00:00.026'. Then, the SQL code for the second table is shown:

```
SQL> CREATE TABLE Loans (  
  LoanID NUMBER PRIMARY KEY,  
  CustomerID NUMBER,  
  InterestRate NUMBER,  
  DueDate DATE  
);
```

Below the code, it states 'Table LOANS created.' The footer of the application shows '26.5.2 - Copyright © 2015, 2026 Oracle and/or its affiliates All rights reserved'. The Windows taskbar at the bottom shows the system clock as 10:16 on 24-06-2026.

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Week 1 Hands-on

2. Insert Data

The screenshot shows the FreeSQL interface with the following SQL script in the editor:

```
1 INSERT INTO Customers VALUES (1,65,15000,'FALSE');
2 INSERT INTO Customers VALUES (2,45,8000,'FALSE');
3 INSERT INTO Customers VALUES (3,70,20000,'FALSE');
4
5 INSERT INTO Loans VALUES (101,1,10,SYSDATE+20);
6 INSERT INTO Loans VALUES (102,2,12,SYSDATE+40);
7 INSERT INTO Loans VALUES (103,3,11,SYSDATE+10);
8
9 COMMIT;
```

The 'Script output' tab is active, showing the execution results:

```
1 row inserted.
Elapsed: 00:00:00.002

SQL> COMMIT

Commit complete.
Elapsed: 00:00:00.002
```

The left sidebar shows the 'My Schema' with tables 'CUSTOMERS' and 'LOANS'. The right sidebar shows the 'SQL Tutorial' community feed.

3. Verify Customers Table

The screenshot shows the FreeSQL interface with the query `SELECT * FROM Customers;` in the editor. The 'Query result' tab is active, displaying the following table:

	CUSTOMERID	AGE	BALANCE	ISVIP
1	1	65	15000	FALSE
2	2	45	8000	FALSE
3	3	70	20000	FALSE

The left sidebar shows the 'My Schema' with tables 'CUSTOMERS' and 'LOANS'. The right sidebar shows the 'SQL Tutorial' community feed.

4. Verify Loans Table

The screenshot shows the FreeSQL interface with the query `SELECT * FROM Loans;` in the editor. The 'Query result' tab is active, displaying the following table:

	LOANID	CUSTOMERID	INTERESTRATE	DUEDATE
1	101	1	10	7/14/2026, 4:50:51
2	102	2	12	8/3/2026, 4:50:51 A
3	103	3	11	7/4/2026, 4:50:51 A

The left sidebar shows the 'My Schema' with tables 'CUSTOMERS' and 'LOANS'. The right sidebar shows the 'SQL Tutorial' community feed.

5. Scenario 1

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

- **Question:** Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

The screenshot shows the SQL Developer interface. On the left, the Navigator pane shows 'My Schema' with tables 'CUSTOMERS' and 'LOANS'. The main editor contains a PL/SQL block:

```
1 DECLARE
2 BEGIN
3   UPDATE Loans
4     SET InterestRate = InterestRate - 1
5     WHERE CustomerID IN
6       ( SELECT CustomerID
7         FROM Customers
8         WHERE Age > 60
9       );
10 COMMIT;
11 END;
```

The 'Script output' tab is active, showing the execution results:

```
UPDATE Loans
SET InterestRate = InterestRate - 1...
Show more...

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.024
```

The footer shows '26.5.2 - Copyright © 2015, 2026 Oracle and/or its affiliates All rights reserved'.

6. Verify Scenario 1

The screenshot shows the SQL Developer interface. The main editor contains the query:

```
1 SELECT * FROM Loans;
```

The 'Query result' tab is active, showing the execution results:

	LOANID	CUSTOMERID	INTERESTRATE	DUE DATE
1	101	1	9	7/14/2026, 4:50:51
2	102	2	12	8/3/2026, 4:50:51 A
3	103	3	10	7/4/2026, 4:50:51 A

The footer shows '26.5.2 - Copyright © 2015, 2026 Oracle and/or its affiliates All rights reserved'.

7. Scenario 2

Scenario 2: A customer can be promoted to VIP status based on their balance.

- **Question:** Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over \$10,000.

The screenshot shows the SQL Developer interface. The left sidebar contains a 'Navigator' pane with 'My Schema' and 'Tables' sections. Under 'Tables', 'CUSTOMERS' and 'LOANS' are listed. The main editor window displays a PL/SQL block in a 'SQL Worksheet':

```
1 DECLARE
2 BEGIN
3   UPDATE Customers
4     SET IsVIP = 'TRUE'
5     WHERE Balance > 10000;
6
7   COMMIT;
8
9 END;
```

Below the editor, the 'Script output' tab is active, showing the execution results:

```
SQL> DECLARE
      BEGIN
        UPDATE Customers
          SET IsVIP = 'TRUE'...
      Show more...

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.014
```

The bottom of the window shows a footer with links: 'About Oracle', 'Contact Us', 'Legal Notices', 'Terms and Conditions', 'Your Privacy Rights', 'Delete Your FreeSQL Account', and 'Cookie Preferences'. The copyright notice reads: '26.5.2 - Copyright © 2015, 2026 Oracle and/or its affiliates All rights reserved'.

8. Verify Scenario 2

The screenshot shows the SQL Developer interface after executing a query to verify the update. The left sidebar is the same as in the previous screenshot. The main editor window displays a simple SQL query:

```
1 SELECT * FROM Customers;
```

Below the editor, the 'Query result' tab is active, showing the results of the query. The execution time is 0.007 seconds. The results are displayed in a table with the following columns: CUSTOMERID, AGE, BALANCE, and ISVIP.

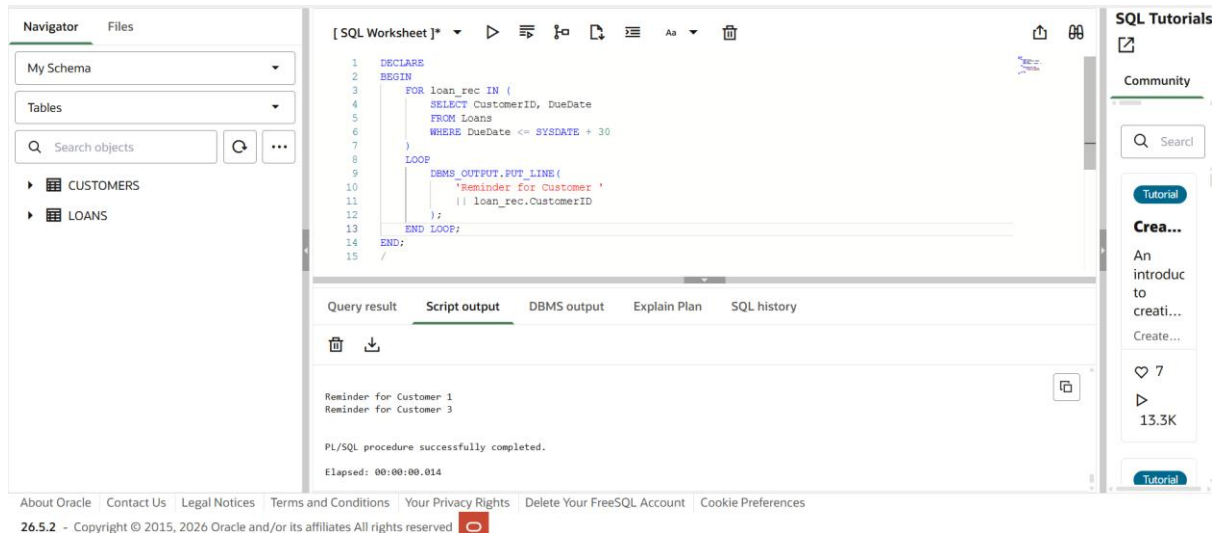
	CUSTOMERID	AGE	BALANCE	ISVIP
1	1	65	15000	TRUE
2	2	45	8000	FALSE
3	3	70	20000	TRUE

The bottom of the window shows the same footer as the previous screenshot, including the copyright notice: '26.5.2 - Copyright © 2015, 2026 Oracle and/or its affiliates All rights reserved'.

9. Scenario 3

Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

- **Question:** Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.



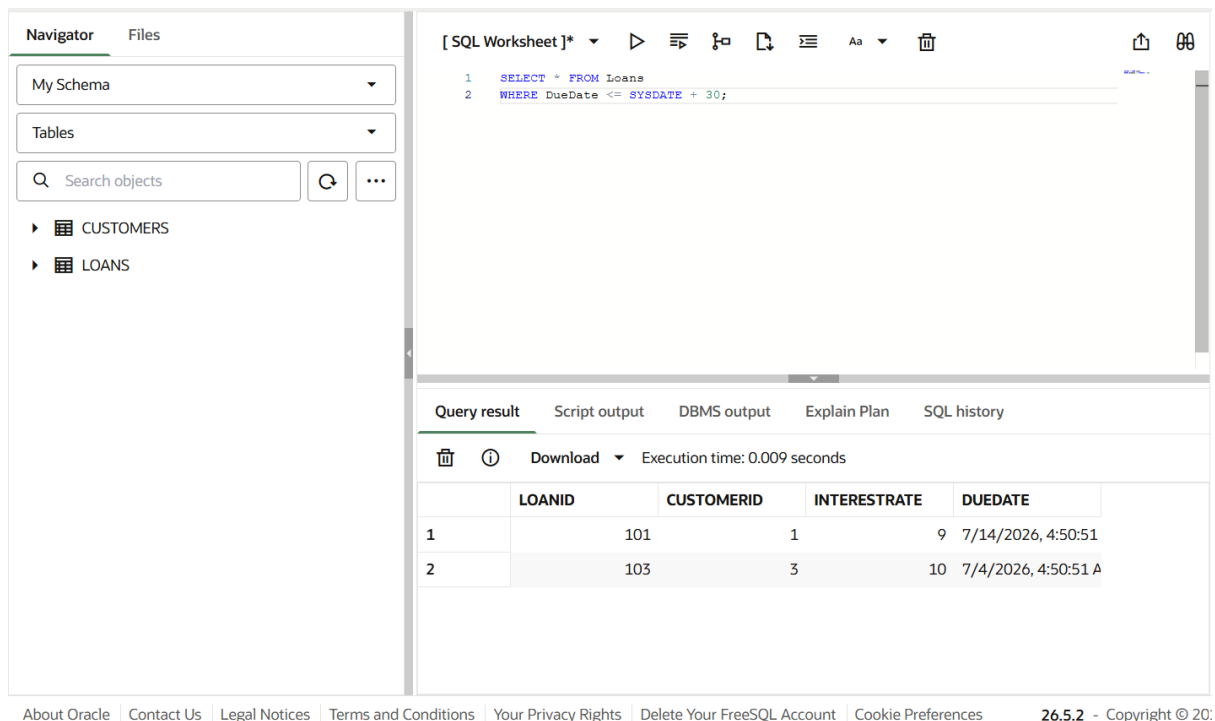
```
1 DECLARE
2 BEGIN
3   FOR loan_rec IN (
4     SELECT CustomerID, DueDate
5     FROM Loans
6     WHERE DueDate <= SYSDATE + 30
7   )
8   LOOP
9     DBMS_OUTPUT.PUT_LINE(
10      'Reminder for Customer '
11      || loan_rec.CustomerID
12    );
13  END LOOP;
14 END;
```

Query result | **Script output** | DBMS output | Explain Plan | SQL history

Reminder for Customer 1
Reminder for Customer 3

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.014

10. Verify Scenario 3



```
1 SELECT * FROM Loans
2 WHERE DueDate <= SYSDATE + 30;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.009 seconds

	LOANID	CUSTOMERID	INTERESTRATE	DUEDATE
1	101	1	9	7/14/2026, 4:50:51
2	103	3	10	7/4/2026, 4:50:51 A

Conclusion

The PL/SQL control structures were successfully implemented.

Scenario 1:

Applied a 1% discount to loan interest rates for customers above 60 years.

Scenario 2:

Updated VIP status for customers with balance greater than \$10,000.

Scenario 3:

Generated reminder messages for customers whose loans are due within the next 30 days.

Thus, all control structure operations were executed successfully.